

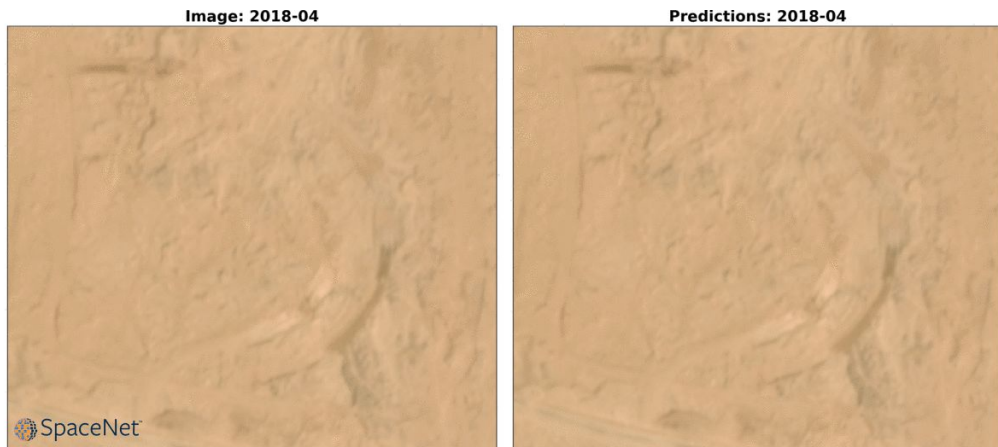


Classification Methods on SpaceNet-7 Satellite Imagery

MATH 7339

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Data - SpaceNet Satellite Imagery



The SpaceNet 7 Multi-Temporal Urban Development Challenge [1, 2]: Participants competed to create the best model to identify building footprints, given two years of satellite imagery from 101 areas

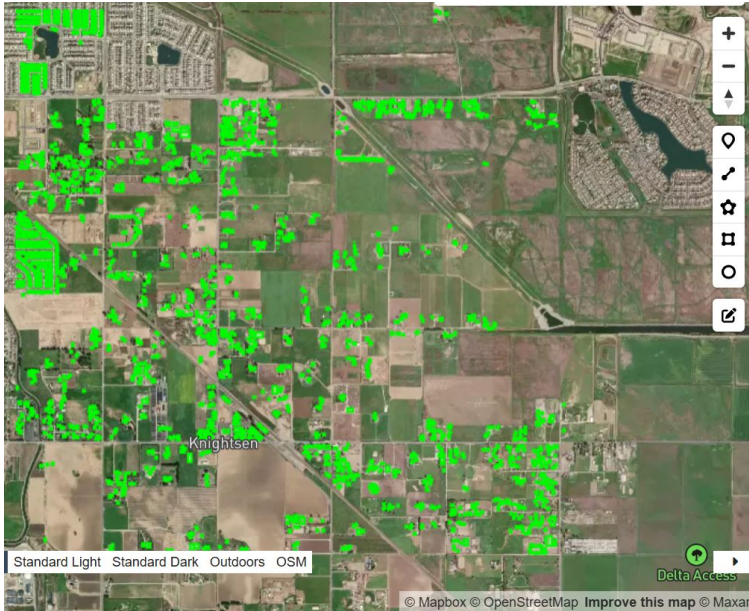
This project shifts focus to classifying **seasons** and **urban sectors**

60 areas of interest (AOI), each has:

- Images (.tif)
 - ◆ Raw and masked
 - ◆ Load using Rasterio into spectral bands for each pixel (e.g. Height, Width, RGBs x image size)
- Labels (.geojson)
 - ◆ Text format for encoding geographical structures
 - ◆ Main labels has one per timestamp
 - ◆ Match labels has all structures with IDs to link over time

For every timestamp (YYYY-MM)

.geojson example



```
1 {  
2   "type": "FeatureCollection",  
3   "features": [  
4     {  
5       "type": "Feature",  
6       "properties": {  
7         "Id": 0  
8       },  
9       "geometry": {  
10        "type": "Polygon",  
11        "coordinates": [  
12          [  
13            [  
14              -121.67141943033411,  
15              37.961579543270666  
16            ],  
17            [  
18              -121.67157154535975,  
19              37.961579543270666  
20            ],  
21            [  
22              -121.67157154535975,  
23              37.961676987182734  
24            ],  
25          ]  
26        ]  
27      }  
28    ]  
29  ]  
30 }
```

Files have real coordinates, used to map and form shapes

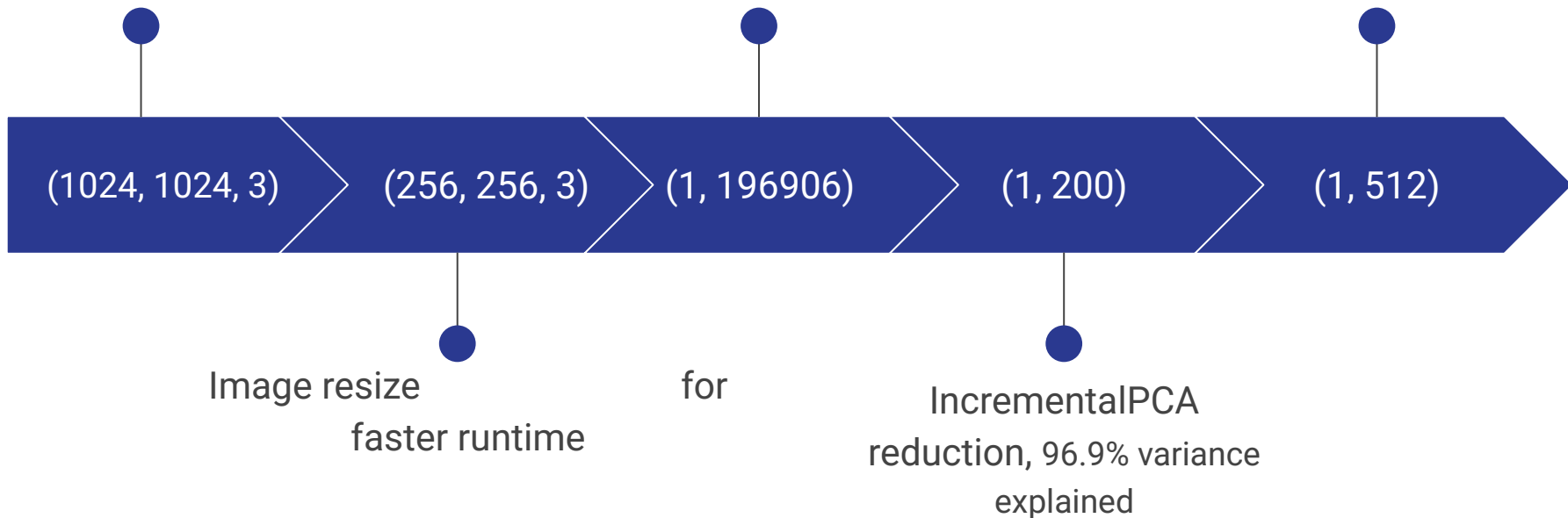
Background is present day map, green marks are from this .geojson from 2018

Image Preprocessing

Original image imported
w/ Rasterio, RGB bands
for each pixel

`np.flatten()`

Alternatively: import image
using Res-Net18 pretrained
CNN [3], get 512 embeddings
per image



Supervised Classifier Models - Season Prediction

Target class: season

Where “winter” -> months 12-2, etc.

Little difference in model performance
(except Naive Bayes) and runtime

Classifier	Accuracy	F1 Score
Bagged RF	0.480702	0.472910
LDA RF	0.473684	0.469493
Random Forest	0.466667	0.464987
KNN	0.466667	0.457261
XGBoost	0.449123	0.447937
SVM	0.435088	0.427811
Naive Bayes	0.266667	0.242460

Large disparity in per AOI model strength



global_monthly_2018_01_mos
aic_L15-0487E-1246N_1950_3
207_13



global_monthly_2018_02_mos
aic_L15-0487E-1246N_1950_3
207_13



global_monthly_2018_05_mos
aic_L15-0487E-1246N_1950_3
207_13

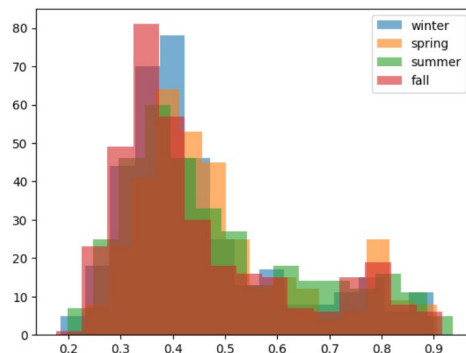
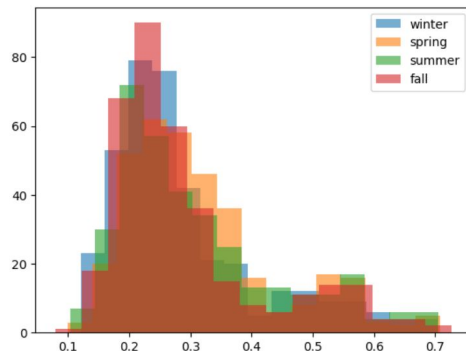


global_monthly_2018_06_mos
aic_L15-0487E-1246N_1950_3
207_13

Some AOIs have perfect predictors, some
have horrible ones

Differences between regions?

Spectral band counts
are indistinguishable



Data is from around
the globe, regional
weather patterns likely
interfere with modeling

Possible to cluster
biomes?

Need to enhance
feature engineering

RandomForest Regressor for “seasonality”

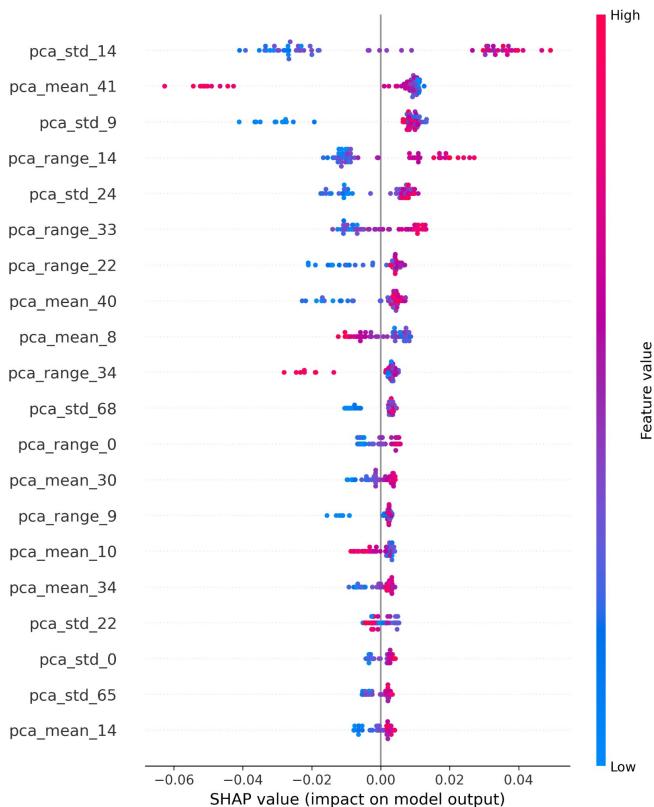
Now using U-Net18 embeddings,
512 -> 75 embeddings w/ PCA, variance
retention of 0.9

Computed AOI features for each:

- For each PCA:
 - ◆ Mean
 - ◆ Standard deviation
 - ◆ Range
- Total variation
- Variation of green

= 60 by $75 \times 3 + 2 = 266$ dataframe

**X = AOI features, y = AOI classifier
accuracies**



Complex Biome Factors

Summer



global_monthly_2018_07_mosai
c_L15-0614E-0946N_2459_4406
_13



global_monthly_2018_08_mosai
c_L15-0614E-0946N_2459_4406
_13

Winter



global_monthly_2018_12_mosai
c_L15-0614E-0946N_2459_4406
_13



global_monthly_2019_01_mosai
c_L15-0614E-0946N_2459_4406
_13

Many more factors to account for than greenness

E.g. sample from AOI 0614E-0946N shows more vibrance in the winter than the summer

-121.67, 37.96 -> Cuzco, Peru; with *wet and dry* seasons

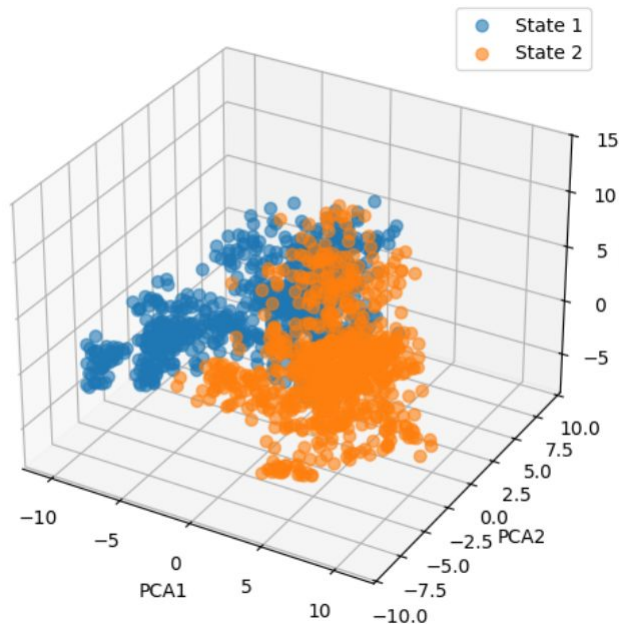
Unsupervised Binary Seasonality (Latent States)

Hard encoding of months to 4 discrete seasons may have limited applicability

Gaussian HMM with two components - general, flexible “does this AOI experience seasons or not”

Some success, still mixing at the middle

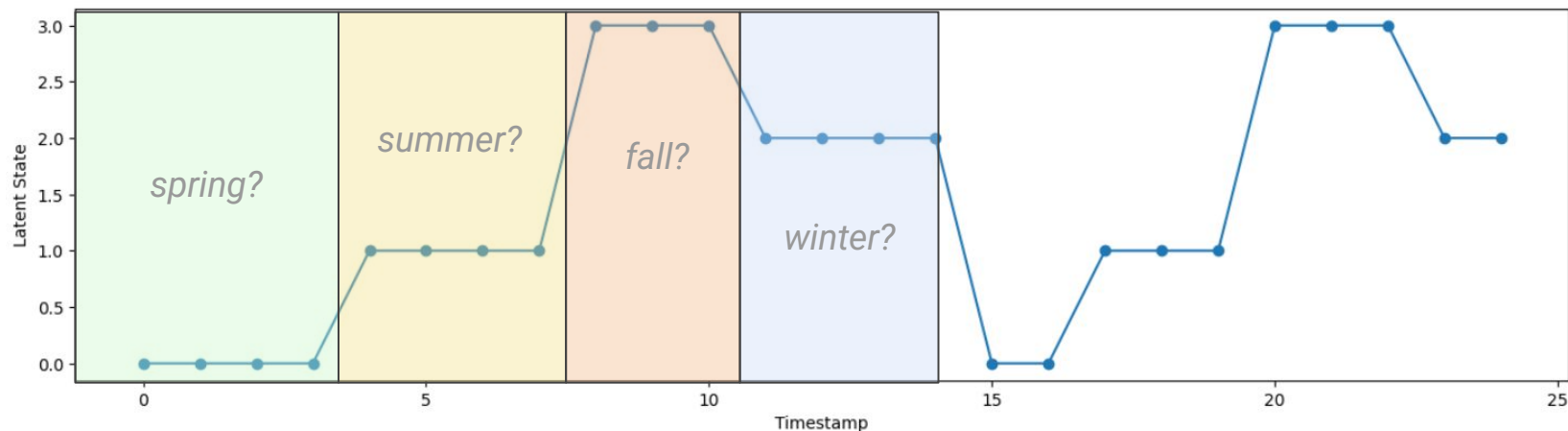
Likely too many combos of regions, biomes, rain patterns, etc. with too little data for clear boundaries



Unsupervised Seasonal States Over Time

When methodology is applied to AOIs with classic visual seasons (e.g. green springs, snow) and western month assignments, latent states do emerge with Gaussian Hidden Markov Models

Example with Contra Costa County, California (outside San Francisco)



Unsupervised Urban Sectors Masking

Question: Can we apply Gaussian Mixture Models to segment land use? (e.g. vegetation, industrial zone, etc.)

Sector Differences

Identify features

relating to sector types
(texture of industrial
roofs, color of vegetation,
etc)

Calculate features

Assemble feature

vectors for each image
patch (e.g. patch 1 has
mean Red band of 0.3, ...)

Fit and plot mask

Fit features

to GaussianMixture(),
view mask overlay,
revalute feature vectors if
needed

Original RGB Image for reference (helps w/ formatting)

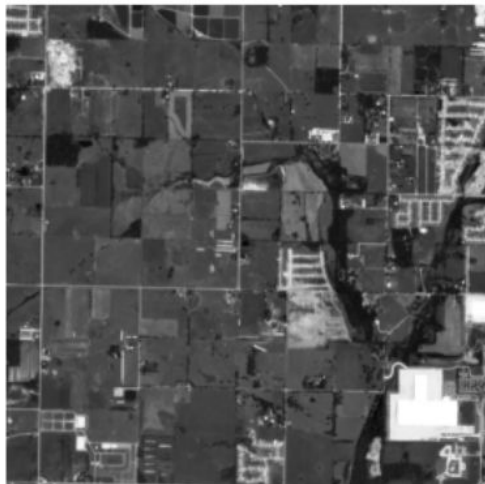


Very Basic Implementation

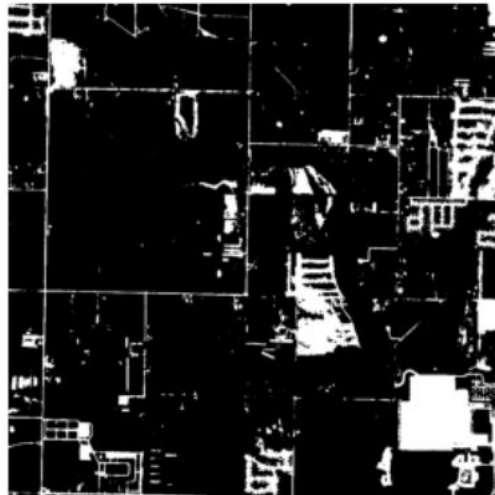
Taking intensity value from each pixel, using GMM to find best threshold to segment image

High intensity pixels of man-made structures makes GMM segmentation effective

Grayscale Image



Binary Classification



First Feature Extracted Mask - Man-made or not

Features calculated for each 8x8 patch:

- Spectrals (mean and std. of each color band)
- Edge density
 - ◆ Sobel library looks for strong changes, convert image to greyscale -> get edges from sobel -> mean() for density

Fit to a 2 component GMM

Man-made vs Natural



Next Feature Extracted Mask - Density

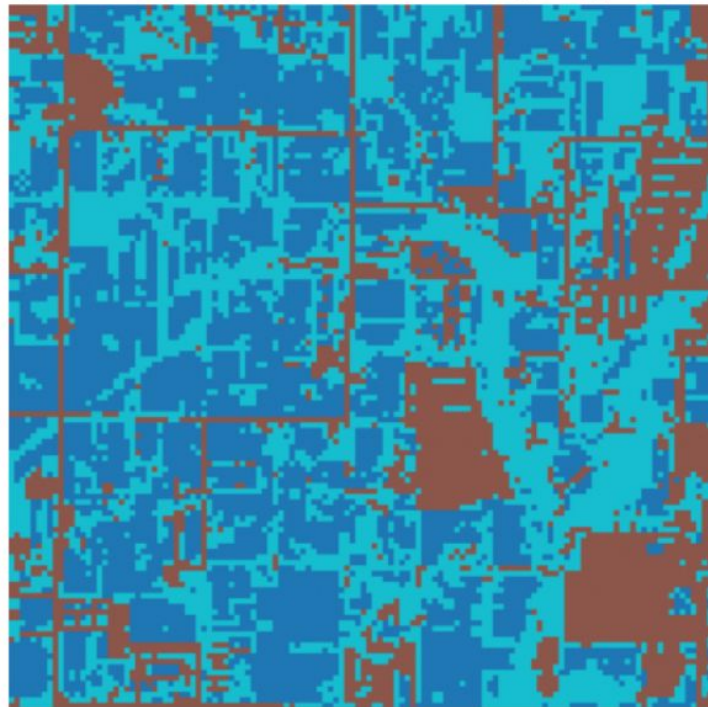
Within the man-made features, which are high vs. low density, e.g. stacked suburbs vs. a scattering of houses

Features added to this one:

- Brightness, greenness, contrast, homogeneity
 - ◆ Skimage.feature functions

Fit to a 3 component GMM

Segmentation: vegetation / low-density / high-density



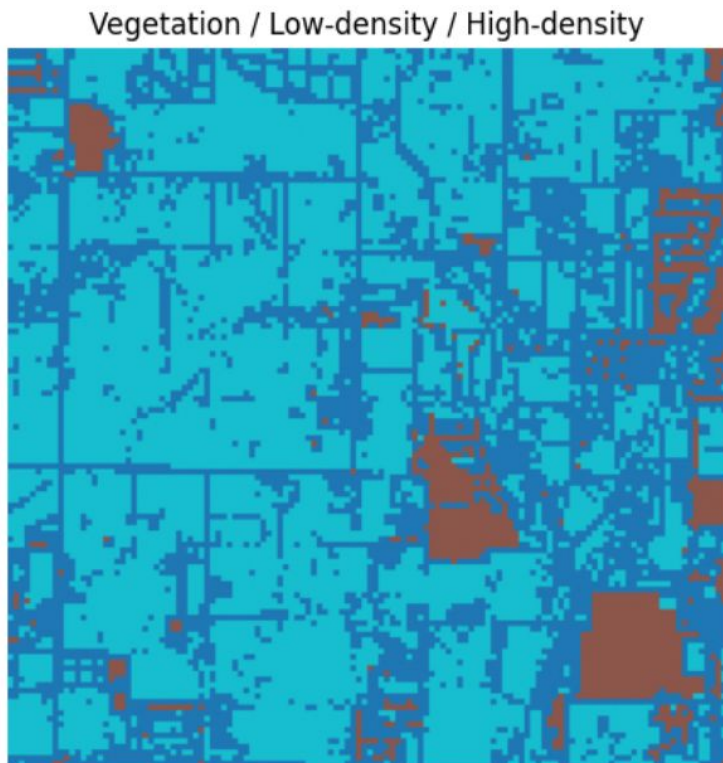
Final Feature Extracted Mask - Enhanced

How to improve feature vector
knowing model is overweighting
color

Features added to this one:

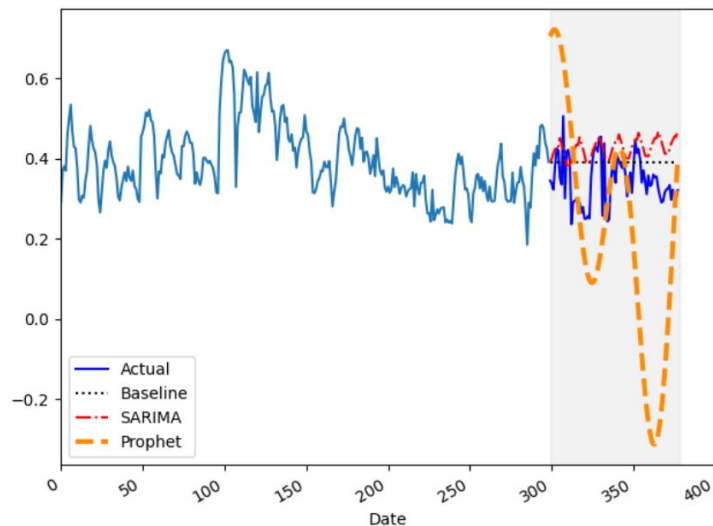
- Corner density
- # of connected components
 - ◆ Both also
skimage.feature
functions

Fit to a 3 component GMM

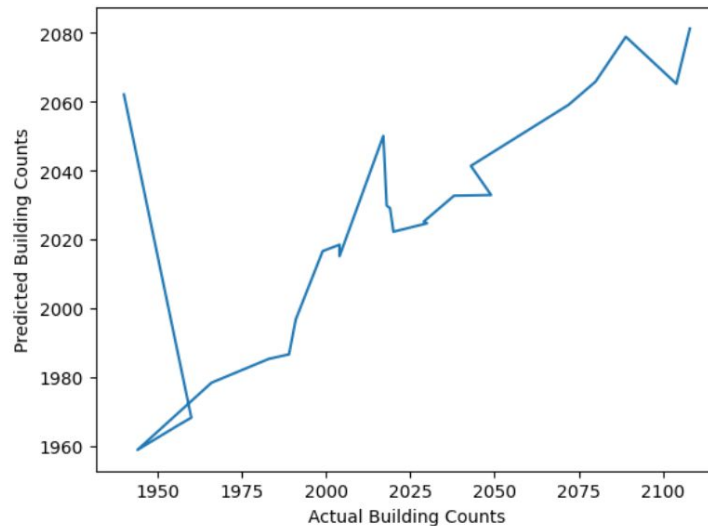


Other Potential Implementations (in early stages)

ARIMA/SARIMA Seasonal Forecasting with mean of green bands per month (many AOIs “stitched” together)



Regression for building count (true labels from length of corresponding .geojson file)



Potential Expansions

Time Series Forecasting:

- Fine-tune ARIMA and SARIMA parameters
- Try different things to model
 - ◆ Sum green bands, building count, development phase, etc.

Building Count Regression:

- Predicted YOY change in development instead of individual buildings

Seasons:

- More time periods available -> stronger models
- More regions available -> opportunity for more robust latent modelling of biomes, rain pattern, uncategorized effects of things like climate change or extreme weather events

Sector Masking:

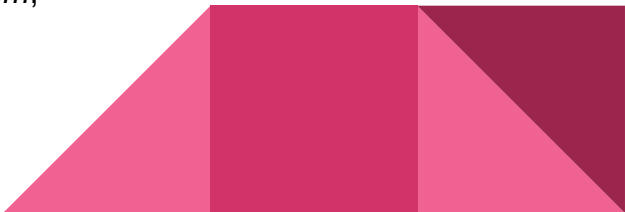
- Mask built for that case, but could be expanded
- Supervised learning opportunity with manually drawn masks

References

Etten, Adam Van. “SpaceNet 7 Results: ArXiv Paper and Winning Model Release.” *Medium*, The DownLinQ, 16 Feb. 2021, medium.com/the-downlinq/spacenet-7-results-arxiv-paper-and-winning-model-release-8da36df16aa5. Accessed 4 Nov. 2025.

Schroeder, Kelly. “Multi-Temporal Urban Development Challenge.” *Spacenet.ai*, 16 June 2020, spacenet.ai/sn7-challenge/. Accessed 4 Dec. 2025.

“ResNet-18 Convolutional Neural Network - MATLAB Resnet18.” *Www.mathworks.com*, www.mathworks.com/help/deeplearning/ref/resnet18.html.



Thank you!

Questions?

